

Introduction to Optimization

Training AI Models

Firuz Kamalov

MAI 600: Mathematics and Statistics for AI

October 24, 2025

- 1 Example: Linear Regression
- 2 Convex vs Nonconvex Functions
- 3 Numerical Optimization

Consider three data points: $\{(0, 1), (1, 2), (2, 1.5)\}$. We seek a line $y = mx + b$ that "best" fits the points in least-squares sense. Define the cost (empirical risk; loss)

$$J(m, b) = \sum_{i=1}^3 (y_i - (mx_i + b))^2.$$

We minimize J over $(m, b) \in \mathbb{R}^2$.

Solve by setting critical points to zero

Compute gradients and set them to zero:

$$\begin{aligned}\frac{\partial J}{\partial m} &= -2 \sum (y_i - mx_i - b)x_i = 0, \\ \frac{\partial J}{\partial b} &= -2 \sum (y_i - mx_i - b) = 0.\end{aligned}$$

For our three points these become a 2×2 linear system.

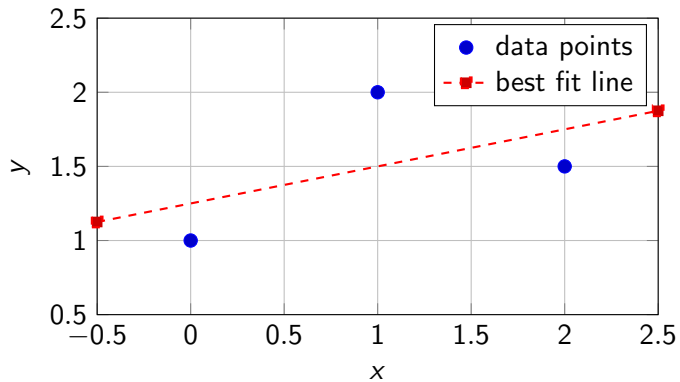
$$5m + 3b = 5$$

$$3m + 3b = 4.5$$

Solving gives

$$m = 0.25, \quad b = 1.25.$$

The minimal cost is $J(m, b) = 0.375$.



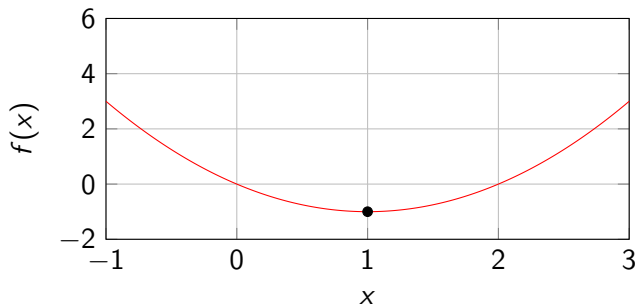
- **Critical point:** $\nabla f(x) = 0$.
- **Local minimum:** $f(x^*) \leq f(x)$ for x in some neighborhood of x^* .
- **Global minimum:** $f(x^*) \leq f(x)$ for all x .
- **Convex function:** $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$ for $\lambda \in [0, 1]$.
- **Hessian criterion:** if $H(x^*)$ positive definite then local minimum; if indefinite then saddle.

Single-variable example: convex

Take $f(x) = x^2 - 2x$. Compute derivative and critical point:

$$f'(x) = 2x - 2, \quad \text{set } f'(x) = 0 \Rightarrow x^* = 1.$$

Second derivative $f''(x) = 2 > 0$ so $x^* = 1$ is a global minimum with value $f(1) = -1$.

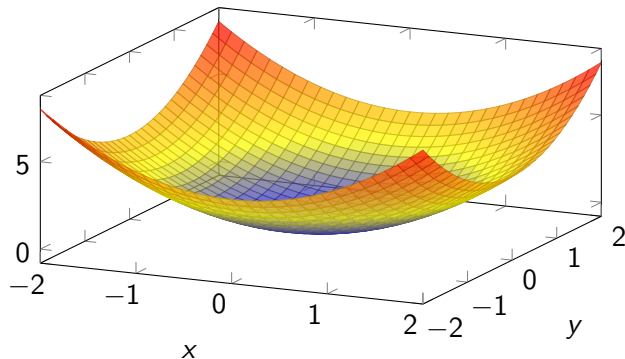


Multivariate convex example

Consider $f(x, y) = x^2 + y^2$. This is convex and has a unique global minimum at $(0, 0)$.

$$\nabla f = \begin{pmatrix} 2x \\ 2y \end{pmatrix}, \quad \nabla f = (0, 0) \Rightarrow (x, y) = (0, 0).$$

Contour plot:



Non-convex example

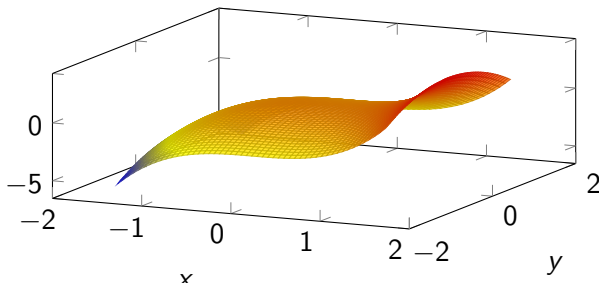
Example: $f(x, y) = x^3 - y^2$.

$$\nabla f = \begin{pmatrix} 3x^2 \\ -2y \end{pmatrix}.$$

Critical points solve $3x^2 = 0$, $-2y = 0$, so $(0, 0)$ is a critical point. However, the Hessian

$$H = \begin{pmatrix} 6x & 0 \\ 0 & -2 \end{pmatrix}$$

evaluated at $(0, 0)$ is 0: so $(0, 0)$ is a saddle point.



- Convex: any local minimum is global. Rich theory & efficient algorithms.
- Nonconvex: may have many local minima and saddle points; global optimum hard.
- Many ML problems (deep nets) are nonconvex; linear models, SVM, logistic regression are convex.

Practical example: SVM (hinge loss)

For labeled data (x_i, y_i) with $y_i \in \{\pm 1\}$, linear SVM solves

$$\min_w \frac{1}{n} \sum_{i=1}^n \max(0, 1 - y_i w^T x_i) + \frac{\lambda}{2} \|w\|^2.$$

The hinge term is convex in w and the regularizer $\|w\|^2$ is strongly convex; the whole objective is convex. Consequence: any local optimizer is global.

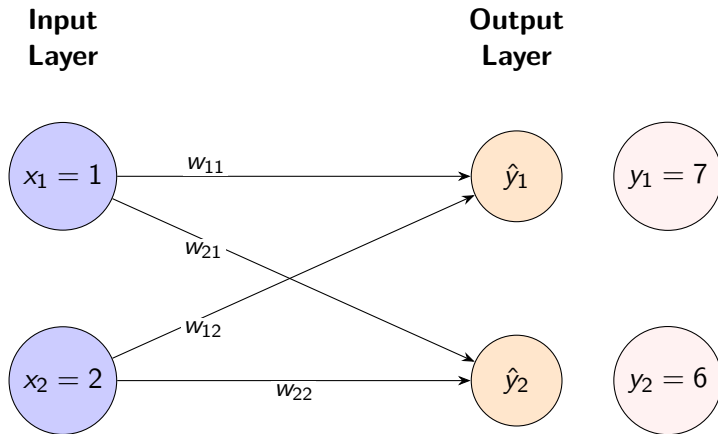
Practical example: small neural network (nonconvex)

Consider a one-hidden-layer network with parameters θ and loss

$$L(\theta) = \frac{1}{n} \sum_{i=1}^n \ell(y_i, f(x_i; \theta)),$$

where f is a composition of linear maps and nonlinear activations. Composition causes nonconvexity in θ in general. Small networks already exhibit multiple local minima and saddle points.

Neural Network Loss Function



$$L(w) = \left(7 - (w_{11} + 2w_{12})\right)^2 + \left(6 - (w_{21} + 2w_{22})\right)^2$$

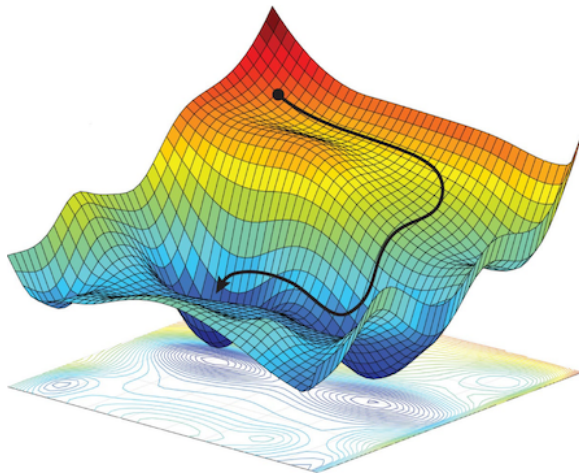
Basic (batch) gradient descent update for minimizing $f(x)$:

$$x_{k+1} = x_k - \eta \nabla f(x_k),$$

where $\eta > 0$ is the step size (learning rate). When f is convex and ∇f is Lipschitz, GD converges with appropriate η .

Gradient descent visualization

Surface of a loss function



https://plus.maths.org/content/sites/plus.maths.org/files/articles/2024/diff_priv/Gradient_descent.gif

Newton update (for twice-differentiable f):

$$x_{k+1} = x_k - H_f(x_k)^{-1} \nabla f(x_k),$$

where H_f is the Hessian matrix. Pros: fast (quadratic) convergence near a nondegenerate minimum. Cons: needs Hessian (costly), may not be positive definite in nonconvex cases.

- Optimization is the mathematical backbone of machine learning.
- Convex problems: well-behaved, global optimality guaranteed.
- Nonconvex problems (deep learning): harder — rely on heuristics and algorithms.
- Understand gradients, Hessians, and algorithmic trade-offs.

Questions?